AI ACADEMY · LEVEL 7 · GRADE 12 · WEEK 5 · WORKBOOK

Replication and reproduction

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

One complete artifact, run and case-evidence record.

Keep data/procedure versions, replay/new status, events, case IDs, outcomes and assumptions together.

Keep one habit: Recall → W1–W3/W5 → independent six-plus-four case → reproduction project → delayed return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Methods results and limitations. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Reading a research claim. Give an example and a mistake that the idea helps you avoid.

Methods results and limitations

> **My remembered idea, example and correction**

Reading a research claim

> **My remembered idea, example and correction**

MAIN LESSON · SECTION 5

W1 · Explain the current example

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Project convention: replay/reproducing computation uses saved artifacts and stated procedure/environment; new-data replication collects new eligible observations under a comparable question/plan. Terms vary, so define them. Main Q1–Q8 original plus same-ID replay gives 16 evaluation events, 8 distinct cases, 0 new cases.

  Different IDs alone do not establish statistical independence, and equal correctness outcomes do not mean duplicate cases. Separate picture recorded correctness flags [1, 0, 1, 1] yield 3/4=75%; same deterministic replay also 75%. New supplied flags [1, 1, 0, 0] yield 2/4=50%; across three runs 12 events/8 distinct IDs/4 new IDs. The picture's independence is a supplied assumption, not tested by IDs.

  New-sample score changes can occur without code error; original artifact unexpectedly scoring 2/4 needs data/code/settings/procedure investigation. Added deliberately changed-analysis example is one possible mismatch mechanism, not a real diagnosis. Preserve versions, protocol and raw fictional outputs. Code executes supplied fixtures, with no actual partner rerun or new collection, and no broad reliability claim.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

My reasoning and evidence

MAIN LESSON · SECTION 6

W2 · Trace the worked case

> Keep the supplied inputs, comparison rule and evidence together.

Complete task and evidence

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- Partner reruns the same eight cases and gets the same score. Team counts sixteen independent cases. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

MAIN LESSON · SECTION 7

W3 · Repair a misconception

> Keep this task's inputs,rule and response together.

Complete task and evidence

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- A report omits this qualification: Exact **replay** can preserve the original design flaws; new data can change conditions and needs its own record.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

W5 · Explain the limitation

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Project convention: replay/reproducing computation uses saved artifacts and stated procedure/environment; new-data replication collects new eligible observations under a comparable question/plan. Terms vary, so define them. Main Q1–Q8 original plus same-ID replay gives 16 evaluation events, 8 distinct cases, 0 new cases.

  Different IDs alone do not establish statistical independence, and equal correctness outcomes do not mean duplicate cases. Separate picture recorded correctness flags [1, 0, 1, 1] yield 3/4=75%; same deterministic replay also 75%. New supplied flags [1, 1, 0, 0] yield 2/4=50%; across three runs 12 events/8 distinct IDs/4 new IDs. The picture's independence is a supplied assumption, not tested by IDs.

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- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

MAIN LESSON · SECTION 4

Try alone · Selected transfer

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Team repeats old six queries, then evaluates four newly reserved queries. How many distinct cases and which are new **evidence**?

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Apply a new case independently

> **My independent explanation, trace and limits**

MAIN LESSON · SECTION 10

Project · Keep the research evidence

> Preserve the question, protocol, observations, analysis and limits.

Complete task and evidence

- Add a labelled **replication** and reproduction **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of **replication** and reproduction.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**
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MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	

Evidence	My record
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Trace the worked case for **Replication** and reproduction. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use the worked case for **Replication** and reproduction. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Team repeats old six queries, then evaluates four newly reserved queries. How many distinct cases and which are new **evidence**?

My reasoning and evidence

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

Extra practice · Pictured case

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Same records versus new experiment. A team first reruns a supplied analysis, then collects new records with the same protocol.

- Original recorded correctness flags: [1,0,1,1]. Analysis counts correct cases. New independent set: [1,1,0,0]. Binary entries mean 1=correct and 0=incorrect.

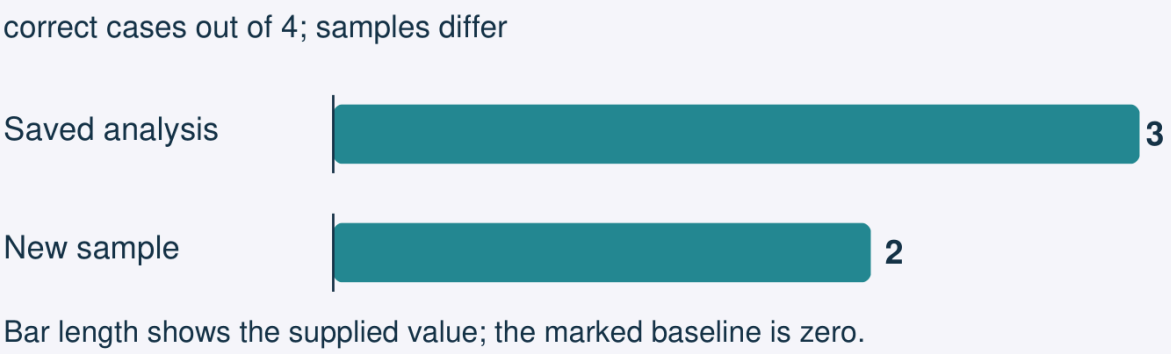
- Calculate the scores from the original and new four-record sets. Which activity reproduces saved analysis, which repeats the protocol, and why does a different rerun of the original data need investigation?

- Support used: alone / hint / worked example. This record is practice **evidence**.
- Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning.

  Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

AI ACADEMY / AI-7/5

Different four-record datasets



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/5

Replication and reproduction

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Reproduce the analysis

Reusing the supplied four records gives 3/4=75%.

2. Repeat the protocol

The new four-record set gives 2/4=50%.

3. Compare purposes

The first checks computational reproduction; the second probes repeatability with new observations.

CHANGE ONE THING

The rerun of the original records outputs 2/4.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Project: Reproducible AI Research Study
- Current milestone: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.
- For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Evidence to keep	My record
This week’s input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Same records versus new experiment. A team first reruns a supplied analysis, then collects new records with the same protocol.
- Which labelled step matches this detail? The first checks computational reproduction; the second probes repeatability with new **observation**s.
- 1. Compare purposes
- 2. Reproduce the analysis
- 3. Repeat the protocol
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Compare purposes The first checks computational reproduction; the second probes repeatability with new **observation**s.

Reusing data and collecting new data answer different questions. A different new-sample result can arise without a code error; inspect both protocols and variability.

Changed-case question and optional hint

- Changed-case practice: The rerun of the original records **output**s 2/4. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. Investigate code, version, **input** or scoring differences.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ Check after the changed attempt

After your attempt, compare with the supported explanation: Investigate code, version, **input** or scoring differences. The same fixed records should give the same deterministic count.

Repair and retain the record

Review Methods results and limitations (ai-7/4). Goal: Explain the **mechanism** and terminology of methods results and limitations.

Short practice: Place the five-photo procedure, 3/5 versus 4/5 finding and missing outdoor **evidence** in their proper categories. Why cannot the result establish that B is best for outdoor farming?.

Return to this week and explain what you changed.

Review Reading a research claim (ai-7/2). Goal: Explain the **mechanism** and terminology of reading a research claim. Short practice: Calculate the baseline and new rates for 2/10 and 4/10.

Explain both twice the rate and the percentage-point change, and say why neither measures every learner’s improvement.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**
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▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

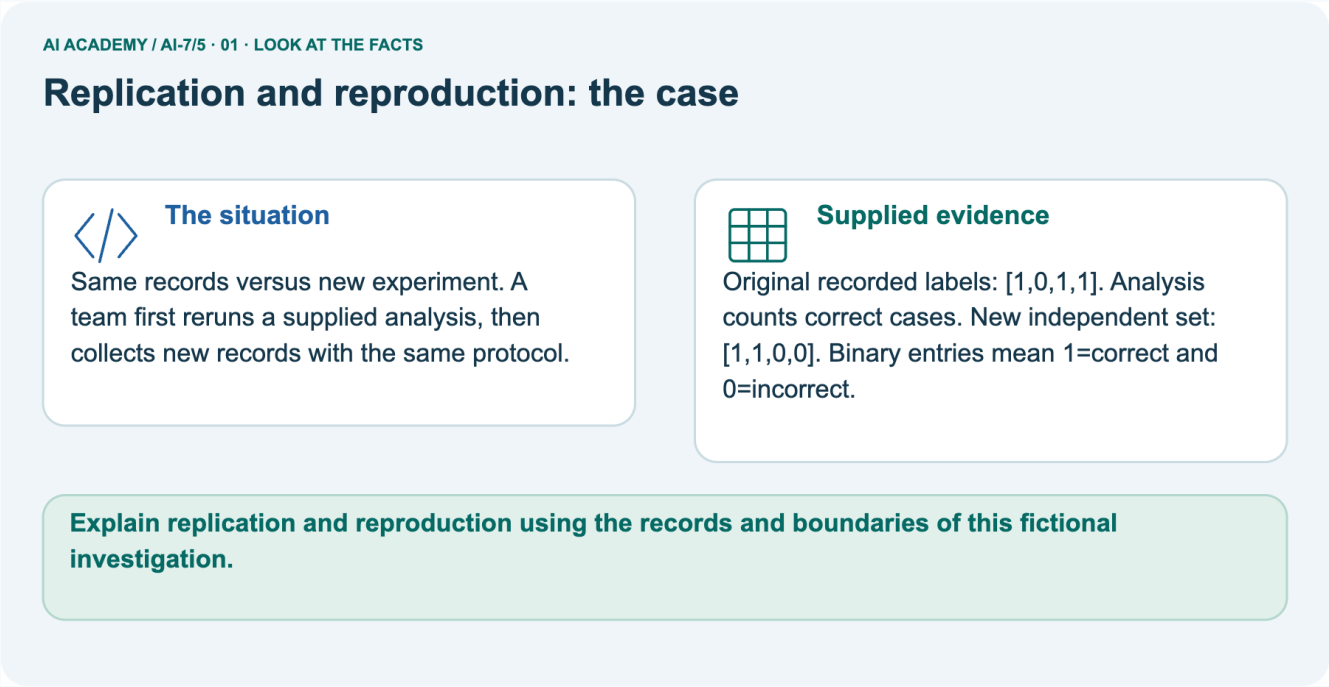
▼ Reference · Six-picture response sheets

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. Replication and reproduction: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

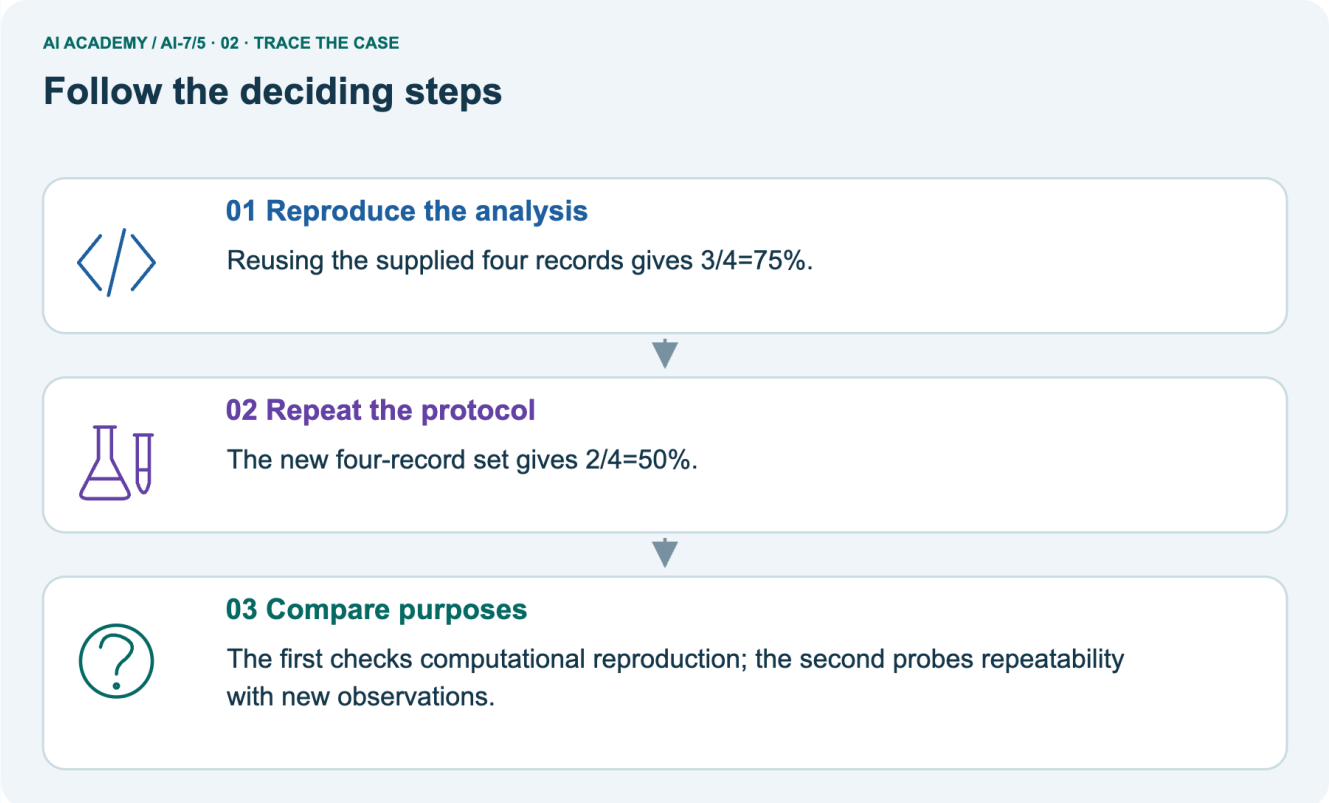
Same records versus new experiment. A team first reruns a supplied analysis, then collects new records with the same protocol. Supplied facts: Original recorded correctness flags: [1,0,1,1]. Analysis counts correct cases. New independent set: [1,1,0,0]. Binary entries mean

1=correct and 0=incorrect.

Point to the given fact that matters most. Explain why it matters.

> **My picture explanation, deciding evidence and one limit**

02. Follow the deciding steps



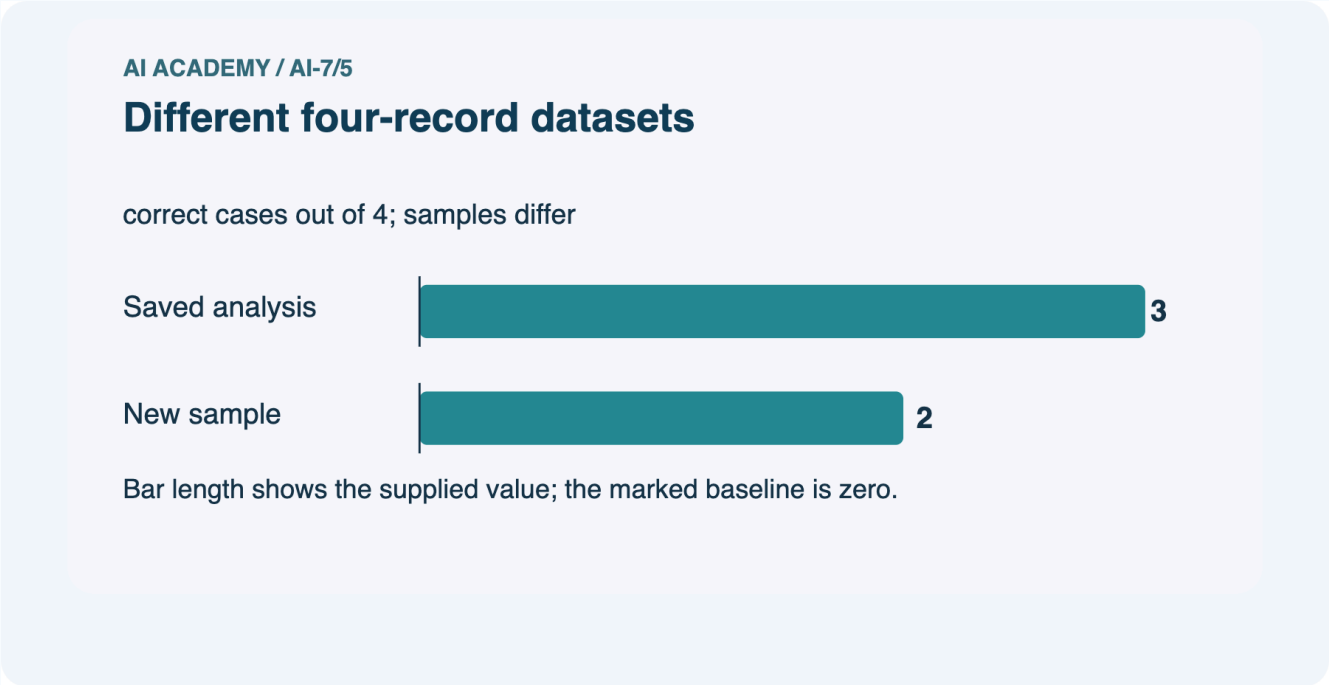
Follow the labelled picture with the matching explanation. Open the vector diagram

Reproduce the analysis: Reusing the supplied four records gives 3/4=75%. → Repeat the protocol: The new four-record set gives 2/4=50%. → Compare purposes: The first checks computational reproduction; the second probes repeatability with new **observation**s.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Different four-record datasets



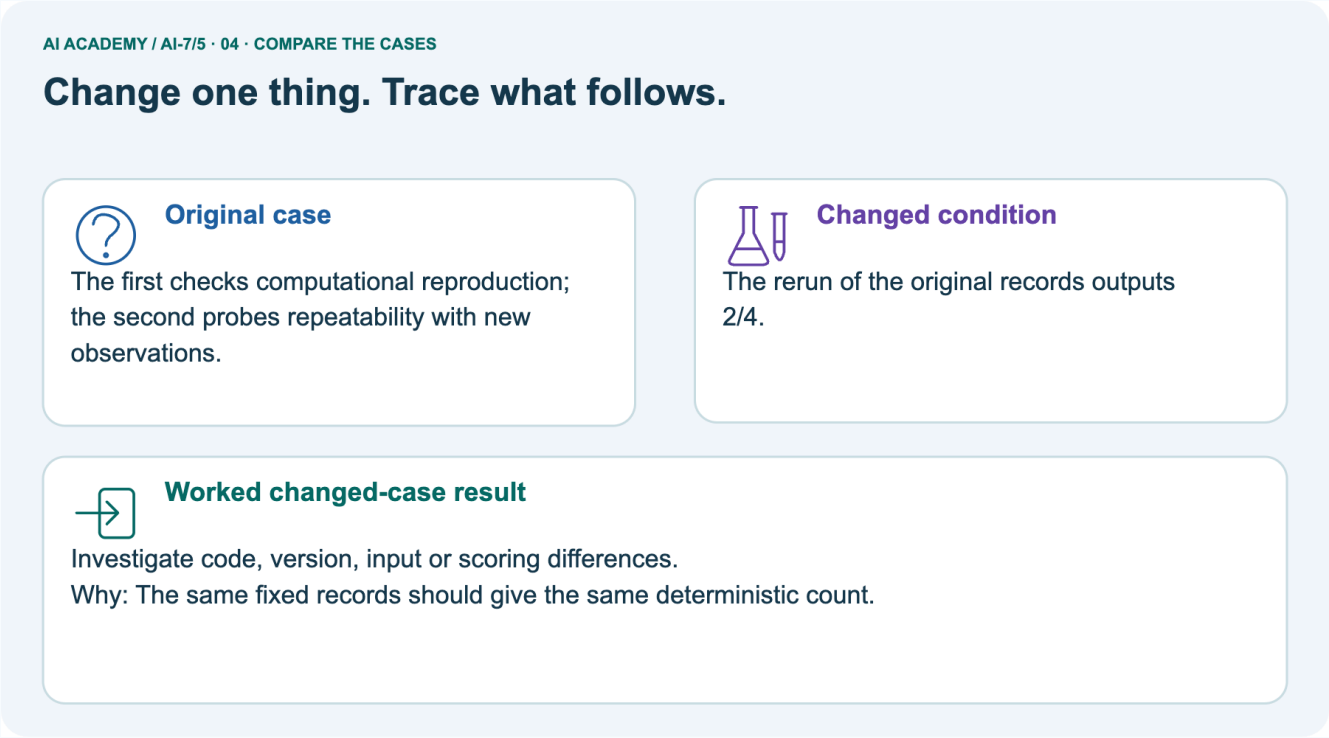
Follow the labelled picture with the matching explanation. Open the vector diagram

Different four-record datasets. Units: correct cases out of 4; samples differ. Saved analysis: 3. New sample: 2.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

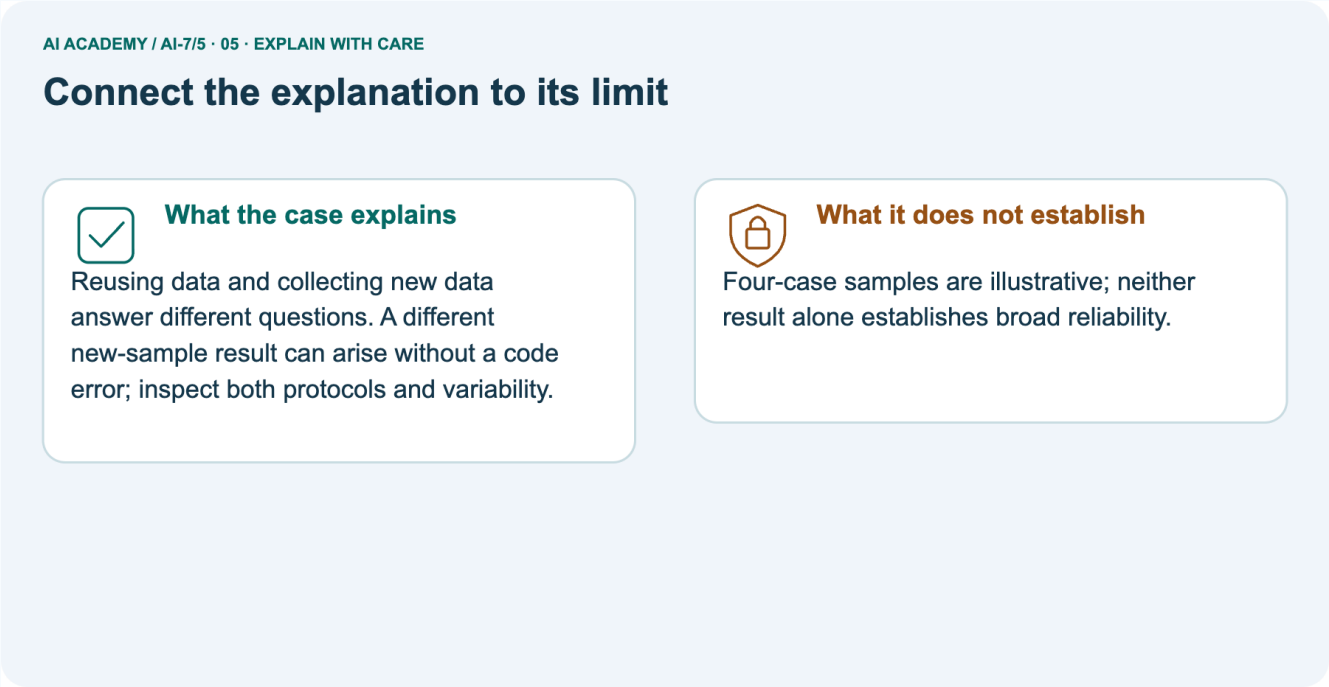
Original case: The first checks computational reproduction; the second probes repeatability with new **observation**s. Changed condition: The rerun of the original records **output**s 2/4.

Expected result: Investigate code, version, **input** or scoring differences. Reason: The same fixed records should give the same deterministic count.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

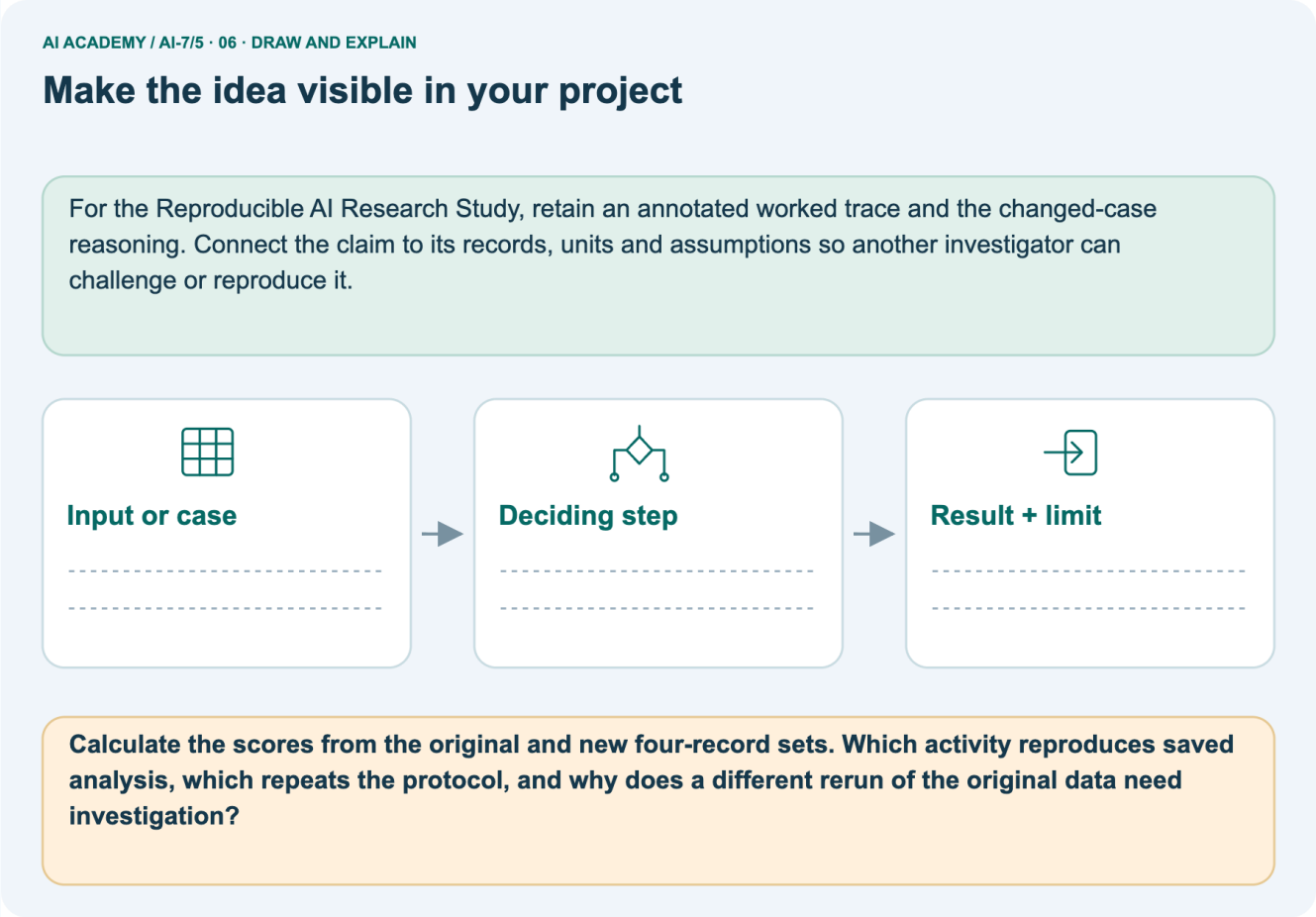
Reusing data and collecting new data answer different questions. A different new-sample result can arise without a code error; inspect both protocols and variability.

Limit: Four-case samples are illustrative; neither result alone establishes broad reliability.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Calculate the scores from the original and new four-record sets.

Which activity reproduces saved analysis, which repeats the protocol, and why does a different rerun of the original data need investigation?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Replication and reproduction: the case

02. Follow the deciding steps

03. Different four-record datasets

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.